

MetabolismGraph: recovering reaction kinetics and homeostatic regulation from metabolite dynamics with graph neural networks

Abstract

A metabolic network is a bipartite graph in which metabolites are coupled through reactions, each with its own rate constant and stoichiometry. We pose the *inverse* problem: given only the time series of metabolite concentrations and the bipartite graph structure, recover the components that generated the dynamics — the per-reaction rate constants k_j , the substrate-to-rate kinetic law, and the per-metabolite homeostatic regulation. We replace the unknown functions of the forward model with learnable MLPs that operate on the metabolite–reaction graph and a vector of learnable rate constants, and train the graph neural network (GNN) end-to-end to predict dc/dt . On synthetic mass-action systems with known ground truth we study rate-constant recovery and the identifiability obstacles that arise from scale ambiguity, functional compensation, and homeostasis absorption. We outline the path from synthetic data to real metabolomics — starting from fully parameterised kinetic models from BioModels (yeast glycolysis), constraint-based genome-scale models (yeast-GEM), and high-frequency real-time metabolomics during nutrient transitions. This is a companion methods note to our connectome-dynamics studies, which apply the same sign-/structure-locked GNN inverse-modelling recipe to neural circuits.

1 Introduction

Cellular metabolism is a large, sparse reaction network: hundreds of metabolites are interconverted by hundreds of enzyme-catalysed reactions, each proceeding at a rate set by an intrinsic rate constant and the instantaneous concentrations of its substrates. The *forward* problem — predicting concentration dynamics given the network, the kinetics and the rate constants — is classical biochemistry. The *inverse* problem is harder and is our subject: given measured concentration trajectories and the graph of which metabolites enter which reactions, recover the rate constants and the functional form of the kinetics. This is the metabolic analogue of the connectome-to-dynamics inverse problems we study elsewhere, and it shares their central difficulty: the mapping from parameters to dynamics is nonlinear, high-dimensional, and degenerate — many parameter combinations produce nearly identical trajectories.

2 Metabolic networks as bipartite graphs

A metabolic network is naturally a **bipartite graph** with two node types: metabolites and reactions. Edges connect only nodes of different type, and each edge carries an integer *stoichiometric coefficient*. A single-partite metabolite–metabolite graph cannot represent the system, because a reaction couples *several* substrates and products at once through a shared rate constant: collapsing it to pairwise metabolite edges would lose the information that those metabolites participate in the same reaction.

Stoichiometric matrix. The connectivity is summarised by the stoichiometric matrix $\mathbf{S} \in \mathbb{R}^{n \times m}$ (n metabolites, m reactions), where S_{ij} is the signed coefficient with which metabolite i enters reaction j : negative for substrates (consumed), positive for products (produced). $\mathbf{S}$ is sparse (each reaction touches only a handful of metabolites), integer-valued (typically in $\{-2, -1, 0, +1, +2\}$), and for mass-balanced reactions its columns sum to zero.

3 Forward model: reaction kinetics and homeostasis

Mass-action kinetics. Under the law of mass action a reaction proceeds faster when its substrates are more abundant, with each substrate raised to the power of its stoichiometric coefficient. For reaction j with rate constant k_j ,

$$v_j = k_j \prod_{k \in \text{sub}(j)} c_k^{|S_{kj}|}, \quad (1)$$

so a reaction consuming two molecules of a substrate depends quadratically on its concentration, and one consuming a single molecule depends linearly. The product over substrates (*multiplicative* aggregation) encodes the requirement that *all* substrates be present simultaneously, and the resulting nonlinear coupling is what allows autocatalytic cycles to sustain oscillations. A simpler *additive* aggregation $v_j = k_j \sum_k c_k^{|S_{kj}|}$ treats substrates as contributing independently and is a useful surrogate for enzyme-mediated reactions that are not strictly mass-action.

Full dynamics. Combining reaction fluxes through $\mathbf{S}$ with a per-metabolite homeostatic restoring force gives the complete forward model

$$\frac{dc_i}{dt} = \underbrace{-\lambda_i (c_i - c_i^{\text{baseline}})}_{\text{homeostasis}} + \underbrace{\sum_{j=1}^m S_{ij} v_j}_{\text{reaction dynamics}}, \quad (2)$$

with v_j from Eq. (1). The homeostatic term pulls each metabolite toward a baseline level c_i^{baseline} at a metabolite-specific rate λ_i ; crucially it is *small* relative to the reaction fluxes, which is what makes it hard to recover (Sect. 5). Eq. (2) spans a family of regimes used for synthetic data: pure reaction ($\lambda = 0$, additive), homeostatic ($\lambda > 0$, additive, convergent), and oscillatory (autocatalytic 3-cycles, multiplicative, sustained oscillations).

4 Inverse problem: the GNN parameterisation

Given observed trajectories $\{c_i(t)\}$ and the bipartite graph, we recover the generating model by replacing every unknown function in Eqs. (1) and (2) with a learnable component on the graph:

$$\frac{dc_i}{dt} = \underbrace{\text{MLP}_{\text{node}}(c_i, a_i)}_{\approx -\lambda_i (c_i - c_i^{\text{baseline}})} + \sum_{j=1}^m S_{ij} k_j \underbrace{\prod_{k \in \text{sub}(j)} \text{MLP}_{\text{sub}}(c_k, |S_{kj}|)}_{\approx k_j c_k^{|S_{kj}|}}. \quad (3)$$

The GNN operates on the metabolite–reaction graph by message passing: substrate states are mapped through MLP_{sub} , aggregated per reaction (multiplicatively for mass-action), scaled by the learnable rate constant k_j , and distributed back to metabolites through $\mathbf{S}$; the node update MLP_{node} adds the

self-regulation term. Training minimises the prediction error on dc/dt end-to-end. The learnable parameters are listed in Tab. 1.

Parameter	Type	Role
$a_i \in \mathbb{R}^d$	embedding vector	per-metabolite identity
k_j	scalar (m of them)	per-reaction rate constant (learned in log-space)
$\text{MLP}_{\text{sub}}(c_k, S_{kj})$	MLP	discovers the kinetic law $c_k^{ S_{kj} }$
$\text{MLP}_{\text{node}}(c_i, a_i)$	MLP	discovers homeostasis $-\lambda_i(c_i - c_i^{\text{baseline}})$

Table 1: Learnable components of the GNN inverse model (Eq. (3)). The stoichiometric matrix $\mathbf{S}$ is either frozen from ground truth (“S given” mode, the focus here) or made learnable (“S learning” mode).

Function discovery. The GNN is not told the kinetic law. MLP_{sub} receives a substrate concentration and its stoichiometric coefficient and must learn the power-law response $c_k^{|S_{kj}|}$ from data alone — in particular it must distinguish linear (c^1) from quadratic (c^2) dependence as the coefficient changes. Likewise MLP_{node} must recover the subtle, small-magnitude homeostatic signal without absorbing dynamics that belong to the reaction terms.

5 Identifiability

The three learnable components interact and can compensate for one another, producing degenerate solutions with high prediction accuracy but poor parameter recovery:

- **Scale ambiguity.** The product $k_j \text{MLP}_{\text{sub}}$ is invariant under $k \rightarrow \alpha k$, $\text{MLP}_{\text{sub}} \rightarrow \text{MLP}_{\text{sub}}/\alpha$. Without anchoring, the MLP absorbs a global scale and shifts every k_j . A centring regulariser pins $\text{mean}(\log k)$ to the known range centre.
- **Functional compensation.** If MLP_{sub} settles on a wrong exponent (*e.g.* $c^{1.5}$ instead of c^2), the rate constants can partly compensate by re-scaling, masking the error.
- **Homeostasis absorption.** MLP_{node} can grow and soak up dynamics that should be explained by reactions, biasing the recovered rate constants.

These are the metabolic counterparts of the sign/scale degeneracies that motivate the structure-locking and regularisation priors in our connectome-dynamics models.

6 Status: one validated working point

It is worth stating plainly where the method stands, because the gap between the synthetic proof-of-concept and real data is large and mostly unbridged. To date there is exactly *one* validated, leak-free working point: recovery of the 256 rate constants k_j on the synthetic oscillatory regime with the stoichiometry $\mathbf{S}$ given. The best single run reaches $R^2 \approx 0.87$ in $\log_{10} k$ (slope ≈ 1 , $\sim 10/256$ outliers), but a cross-seed reproduction puts the honest working point lower, at $R^2 = 0.75 \pm 0.04$ (Sect. 7). This is a genuine result — the loss reads only the concentration dynamics, never any ground-truth label.

Two caveats bound even this point. First, an earlier and more favourable “homeostasis recovery” result was *discarded*: it relied on a supervised contrastive term that read ground-truth metabolite-type

labels during training (label leakage), and once that term was removed the homeostatic function MLP_{node} no longer separates — it stays flat, because single-step dc/dt training cannot see a force that only acts over many steps. Recovering homeostasis honestly (a two-phase / residual scheme, Eq. (2)) is unsolved. Second, the k_j recovery itself carries a run-to-run variance of $\pm 0.2\text{--}0.3 R^2$ at fixed hyperparameters, so the working point is a *distribution*, not a fixed number, and reducing that variance is open. Everything beyond rate-constant recovery on synthetic data — homeostasis, stoichiometry learning, and every real-data rung below — remains to be done.

7 Reproduction and cross-seed robustness

The single validated point (Sect. 6) was reproduced from scratch on a fresh data generation, then cross-validated across training seeds to ask how much of the headline $R^2 \approx 0.87$ is reproducible versus a favourable draw. All runs use the identical leak-free configuration (**k_recovery_winner**: rank-50 oscillatory, **S** given, **aggr** = mul, k -floor and MLP_{sub} -norm priors), varying only the optimisation seed. Tab. 2 reports the recovered rate-constant R^2 .

Seed	training completed	k recovery R^2
77 (published winner)	100%	0.742
42	100%	0.705
79	90%	0.808
7	90%	0.738
<i>mean $\pm$ s.d. (4 mature seeds)</i>		0.748 $\pm$ 0.037
123	30% (stopped)	0.722 (rising)

Table 2: Cross-seed reproduction of the leak-free k_j -recovery working point. Four seeds run to (near-)completion cluster tightly at $R^2 = 0.75 \pm 0.04$; seed 123 was stopped early (to free GPUs for the real-data runs) at 0.72 and still climbing. The runs were deliberately halted once the cross-seed picture was clear.

Read. The working point reproduces *robustly* — the cross-seed spread is small (± 0.04), not the ± 0.2 that single-run exploration logs had suggested — but it reproduces at $R^2 \approx 0.75$, not 0.87. The published 0.87 was the best of many draws; the honest, reproducible recovery of the 256 rate constants is 0.75 ± 0.04 .

Fig. 1 unpacks what that 0.75 means. The seed-77 run recovers the rate constants with *raw* $R^2 = 0.74$ over all 256 reactions, but a *trimmed* $R^2 = 0.98$ once the small outlier tail is removed: 245/256 reactions sit on the identity line at unit slope, and only 11 (4.3%) are mis-estimated by more than 0.3 dex. The raw R^2 is therefore dragged down by a few hard reactions, not by a diffuse failure. We adopt a hard quality gate — the outlier fraction must not exceed 10% — which this run passes comfortably (4.3%). This raw-versus-trimmed split is the number to carry into the real-data comparisons.

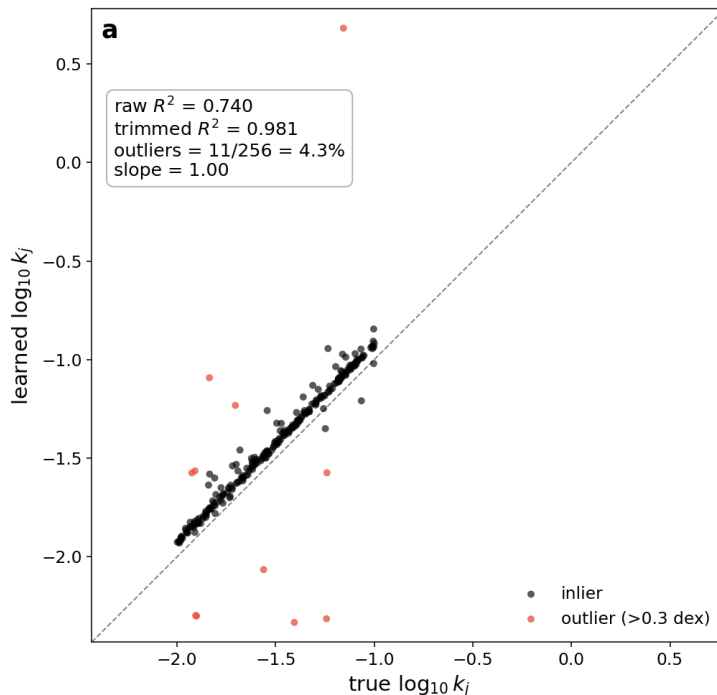


Figure 1: **Toy-model rate-constant recovery (leak-free, seed 77).** Learned vs. true $\log_{10} k_j$ for all 256 reactions after the MLP_{sub} scalar (α) correction; dashed line is identity. Inliers (black) lie on the diagonal at slope 1.00; the 11 red points (4.3%) exceeded the 0.3-dex outlier threshold. Raw $R^2 = 0.74$ over all reactions, trimmed $R^2 = 0.98$ excluding outliers, within the 10%-outlier gate. Reproduced by `figures/k_recovery.py`.

8 Toward real metabolomics

The framework is developed and validated on synthetic data where ground truth is known; the goal is real metabolomic time series. Three concrete datasets, now vendored in the repository, stage that transition from fully parameterised kinetics, through genome-scale topology, to a real recording. None of the three is solved yet; this section describes what each provides and what stands between it and a result.

Rung 1 — yeast glycolysis kinetic model (BioModels MODEL1303260010). A small, fully parameterised kinetic model of glycolysis in *S. cerevisiae* (Smallbone et al., 2013), imported from SBML. After separating dynamic from boundary species it gives 20 internal metabolites (+9 fixed boundary species), 30 reactions including iso-enzymes (three hexokinase, three pyruvate-decarboxylase, three glyceraldehyde-3-phosphate variants), with real stoichiometry $\mathbf{S}$ and measured initial concentrations. Crucially the rate laws are *not* mass-action: they mix reversible/irreversible Michaelis–Menten, Hill, and allosteric forms, so $v_j = V_{\max,j} \prod_k [c_k / (K_{m,kj} + c_k)]^{|S_{kj}|}$ rather than a power law. This is the cleanest available ground truth: known $V_{\max}/K_m$ let us score parameter recovery directly. *Status*: the SBML import, $\mathbf{S}$ extraction, and a Michaelis–Menten simulator are in place and generate a 12,000-frame dataset; a first GNN fit does *not* yet recover the dynamics (autoregressive rollout diverges), so the open problem is making MLP_{sub} learn a saturating law and the homeostatic clamp

behave under a stiff, stimulus-driven system.

Rung 2 — consensus genome-scale model (yeast-GEM). The community-curated genome-scale model of *S. cerevisiae* (Lu et al., 2019): 2,806 metabolites, 4,131 reactions across 14 compartments. It supplies a large, real stoichiometric matrix $\mathbf{S}$ and flux bounds but *no* kinetic rate laws — the true inverse problem. The intended use is to extract a central-carbon subgraph (glycolysis + TCA + fermentation, ~ 100 – 150 reactions), drive it with dynamic-FBA or literature kinetics to generate in-silico dynamics, and fit the GNN in “S given” mode on a realistic topology. *Status*: not started beyond vendoring the model; subgraph extraction and a dynamics generator are still to be written.

Rung 3 — real-time metabolomics during nutrient transitions (real data). High-frequency flow-injection mass spectrometry of live microbial culture (Link et al., 2015), now downloaded as the paper’s supplementary tables. For *E. coli* the released data give a metabolite $\times$ time intensity matrix of 247 annotated ions over 119 evenly spaced time points in three biological replicates (a long batch course), plus a dedicated starvation $\rightarrow$ growth switch series of 302 ions over ~ 148 points; companion *B. subtilis* and *Y. lipolytica* sets are included. Each ion carries a KEGG annotation (444 distinct KEGG IDs for *E. coli*), though many ions are mass-ambiguous (one m/z maps to several candidate metabolites) and values are *relative* ion intensities, not absolute concentrations. *Status*: the data are in hand but not yet ingested; the work is to deduplicate/annotate ions, build $\mathbf{S}$ for the measured subset from an *E. coli* genome-scale model (*e.g.* iJ01366) via KEGG $\rightarrow$ BiGG mapping, and rescale the relative intensities. There is *no* kinetic ground truth here, so success can only be judged by held-out prediction, not by recovered k_j . A parallel mammalian direction is chronic jugular microdialysis in freely moving mice (Nardin et al., 2025) (~ 123 metabolites every 7.5 min over ~ 8 h; 10 components explain 73% of variance).

What transfers and what changes. The core recipe transfers: the bipartite graph, message passing over reactions, function discovery via $\text{MLP}_{\text{sub}}/\text{MLP}_{\text{node}}$, and log-space rate-constant recovery. Real data adds, all at once, the obstacles the synthetic setup omits: non-mass-action (saturating, allosteric) kinetics that MLP_{sub} must learn; a partially known and incomplete $\mathbf{S}$ (only measured metabolites, ambiguous identities); two orders of magnitude fewer, unevenly spaced time points ($\sim 10^2$ vs $\sim 10^3$); relative rather than absolute concentrations; and external drivers (the nutrient switch) that must enter as a term $f_{\text{ext}}(t)$ in Eq. (2).

What we expect. We do *not* expect the real-data rungs to work soon. The realistic near-term target is Rung 1: get the GNN to recover the published V_{max}/K_m of a real kinetic model under Michaelis–Menten kinetics — a contained problem with ground truth, but one the current code does not yet solve. Rung 2 is a topology-realism check with no kinetic ground truth. Rung 3 (real *E. coli*) is the genuine goal but also where every difficulty above appears together; a first “result” there means only predicting held-out frames of measured dynamics, not recovering mechanism. Tab. 3 summarises the gap. The honest reading: one validated synthetic working point, three real rungs at the data-and-scaffolding stage, and substantial method development (saturating kinetics, honest homeostasis, incomplete-network handling, relative-concentration rescaling) still required before any real-data claim.

Target	Network / kinetics	Ground truth	Status
Synthetic k_j (S given)	random / mass-action	k_j known	works , $R^2 \approx 0.87$
Synthetic homeostasis	random / mass-action	λ_i known	open (single-step blind)
Rung 1: yeast glycolysis	real / MM+Hill	$V_{\max}, K_m$ known	data ready; fit fails
Rung 2: yeast-GEM subset	real / none (FBA)	none	not started
Rung 3: real <i>E. coli</i>	inferred / unknown	none	data in hand; not ingested

Table 3: Where each target stands. Only the first row is a validated, leak-free result; the rest is the work ahead.

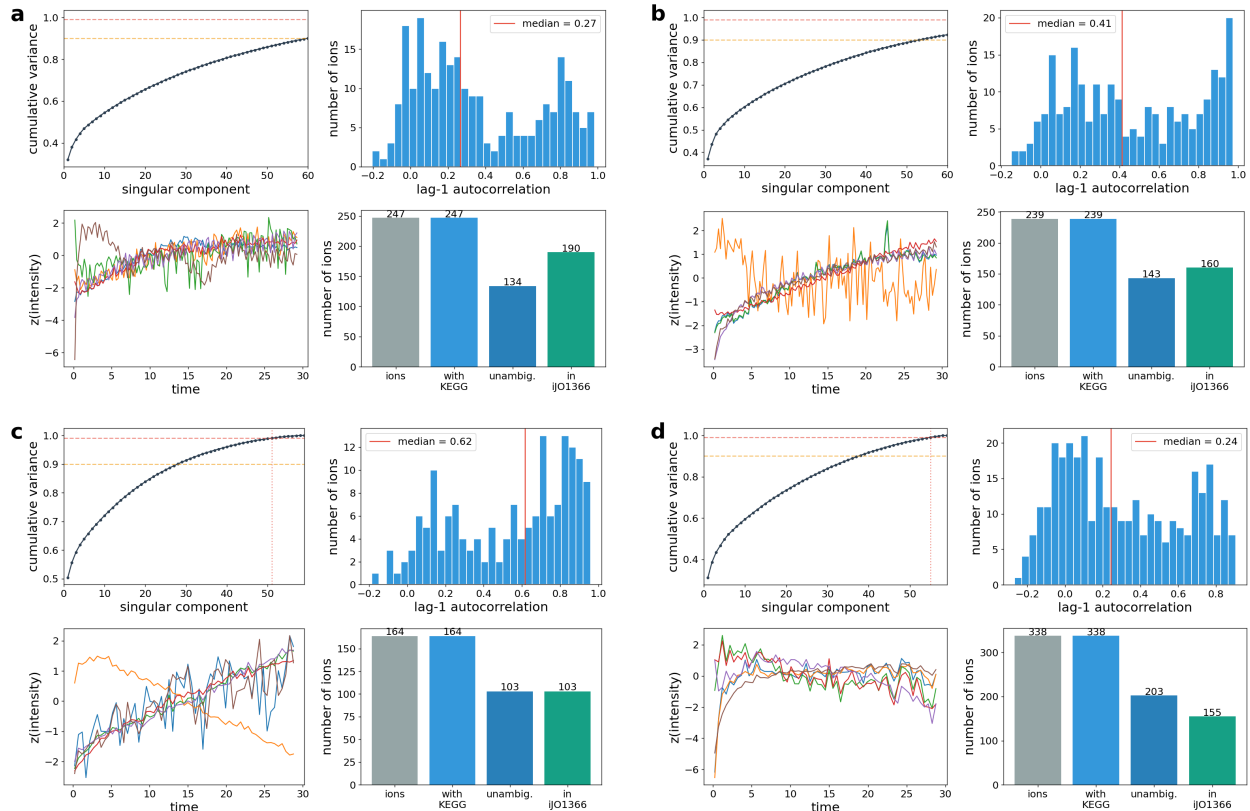
8.1 First look: are the four real datasets amenable?

Before training, we run the same cheap test on *all four* datasets measured by Link et al. (2015) — *E. coli*, *B. subtilis*, *Y. lipolytica*, and a hybrid sample — asking whether each recording carries the structure the GNN needs. The identical four-probe analysis (`figures/amenability.py`) is applied to every organism: SVD low-rank, per-ion lag-1 autocorrelation, example traces, and KEGG coverage against a common reference network (`iJO1366`; a proxy for core-metabolite mappability — exact for *E. coli*, approximate for the others). Tab. 4 summarises and Fig. 2 shows the per-organism panels.

Organism	ions $\times$ t	rank ₉₉	autocorr med.	in iJO1366	dc/dt
<i>E. coli</i>	247 $\times$ 119	102 (41%)	0.27	190 (77%)	noisy
<i>B. subtilis</i>	239 $\times$ 119	98 (41%)	0.41	160 (67%)	noisy
<i>Y. lipolytica</i>	164 $\times$ 59	51 (31%)	0.62	103 (63%)	smooth
Hybrid	338 $\times$ 60	55 (16%)	0.24	155 (46%)	noisy

Table 4: Amenability of the four measured datasets. rank₉₉ is the number (and fraction) of singular components for 99% variance; autocorr is the median per-ion lag-1 autocorrelation; coverage is the count mapped onto `iJO1366`. All four are low-rank and reasonably network-mappable; only *Y. lipolytica* has smooth (dc/dt -learnable) dynamics.

Does it generalise? Two of the three signals hold across all four datasets: the dynamics are *low-rank* (rank₉₉ = 16–41% of ions) and a large share of metabolites are core/conserved (coverage 46–77%). The third — temporal smoothness of dc/dt — does *not* generalise: *E. coli*, *B. subtilis* and the hybrid are noise-dominated (median lag-1 autocorrelation 0.24–0.41), whereas ***Y. lipolytica* is smooth** (median 0.62, 59% of ions above 0.5). The actionable consequence: the most promising target for an actual real-data fit is *not E. coli* (best network coverage but noisiest) but ***Y. lipolytica*** — smooth, learnable dynamics that are still low-rank and network-mappable. The hybrid is least amenable (lowest coverage, noisy), though most strongly low-rank. A real fit should therefore start from *Y. lipolytica*, restricted to its smooth, high-variance, network-mapped ions and judged by held-out prediction — not by mechanism recovery across the whole noisy measurement.



Data and code

Synthetic generators, the GNN model, and the LLM-driven exploration loop are in the `MetabolismGraph` repository; an overview minisite is at <https://allierc.github.io/MetabolismGraph/>. The reference material is vendored under `papers/`: the two yeast models, BioModels MODEL1303260010 (<https://github.com/biomodels/MODEL1303260010>) and yeast-GEM (<https://github.com/SysBioChalmers/yeast-GEM>); and the Link et al. (2015) real-time-metabolomics supplementary tables under `papers/nmeth3584/`. The Rung-1 SBML→dynamics bridge is the `glycolysis_yeast` config/dataset; the leak-free synthetic k_j -recovery working point is the `k_recovery_winner` config.

A Weekend hypothesis ledger (live)

This appendix is updated autonomously every few hours during unattended runs. Each row is a hypothesis with its experiment and a *validated* / **FALSIFIED** / *in progress* verdict. Falsifications are recorded as plainly as positive results.

Date	Hypothesis	Experiment	Verdict
2026-06-05	H1: winner reproduces k $R^2 \approx 0.87$	seed 77, leak-free, S given	0.74 raw / 0.98 trim (4.3% out, P.
2026-06-05	H2: k R^2 robust across seeds	CV 4 seeds, S given	<i>validated</i> (0.75 ± 0.04)
2026-06-05	single-step rollout fits glycolysis	glycolysis_yeast, $t \rightarrow t+1$	FALSIFIED ($R^2 = -390$)
2026-06-05	H4: AR curriculum fixes the rollout	opt-in n -steps curriculum	sweep running (glyco_ar_base)
2026-06-05	H5: real <i>E. coli</i> is low-rank	SVD of 247×119 matrix	<i>validated</i> (rank ₉₉ =102, 41%)
2026-06-05	H6: real dc/dt is learnable/smooth	per-ion lag-1 autocorrelation	FALSIFIED (med 0.27, noisy)
2026-06-05	H7: ions map to a real network	KEGG $\cap$ iJO1366 / e_coli_core	<i>validated</i> (190/247 in iJO1366)
2026-06-05	generalises to 4 organisms?	same probes: Bsubt/Ylipo/hybrid	low-rank & cover <i>yes</i> ; smooth <i>no</i>
2026-06-05	best real fit target	cross-organism dc/dt smoothness	<i>Y. lipolytica</i> (autocorr 0.62)

Table 5: Live experiment ledger. The single-step rollout on the real Michaelis–Menten glycolysis model diverges ($R^2 = -390$, Pearson 0.19), motivating the autoregressive curriculum (Sect. 6) ported from the connectome models: per-epoch horizon ramp $10 \rightarrow 300$, soft tail-loss, LR co-ramp, gradient clipping. Implementation is in place and smoke-validated (runs, finite loss); the 5-config sweep is queued behind the CV runs.

References

- Smallbone, K., Messiha, H.L., Carroll, K.M., *et al.* (2013). A model of yeast glycolysis based on a consistent kinetic characterisation of all its enzymes. *FEBS Letters* 587(17), 2832–2841. BioModels MODEL1303260010.
- Lu, H., Li, F., Sánchez, B.J., *et al.* (2019). A consensus *S. cerevisiae* metabolic model Yeast8 and its ecosystem for comprehensively probing cellular metabolism. *Nature Communications* 10, 3586. yeast-GEM, SysBioChalmers.
- Link, H., Fuhrer, T., Gerosa, L., Zamboni, N., & Sauer, U. (2015). Real-time metabolome profiling of the metabolic switch between starvation and growth. *Nature Methods* 12, 1091–1097. <https://www.nature.com/articles/nmeth.3584>.
- Nardin, M., Wang, N., Elziny, S., *et al.* (2025). Reconstructing a physiological state space via chronic jugular microdialysis in freely moving mice. *bioRxiv*. <https://doi.org/10.64898/2025.12.08.692974>.